Heterogeneous Bipartite Graph Neural Networks for Lattice Quantum Field Theory

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We introduce a heterogeneous bipartite graph neural network (GNN) architecture that separates *spacetime geometry* from *field content* into distinct node types connected by typed edges—a design inspired by the geometry–matter distinction in continuum quantum field theory. Unlike existing approaches that embed all information on spacetime nodes in a homogeneous graph, our architecture represents fields as a separate layer coupled to the lattice through typed "inhabits" edges and three-stage message passing. We benchmark on two-dimensional scalar ϕ^4 theory in the Ising universality class ($\nu = 1$), comparing against homogeneous GNN, convolutional, and multilayer-perceptron (MLP) baselines. We show that homogeneous GNN representations suffer a structural failure mode—message passing destroys the field signal without ad hoc skip connections—which the bipartite design eliminates by construction. The model achieves near-perfect action prediction across lattice sizes ($r > 0.9999$ at 8×8 and 16×16; median $r = 0.9992$ at 64×64), and a single model trained at 16×16 transfers to lattices from 8×8 to 64×64 with $r = 1.0000$ and no retraining. The underlying Monte Carlo pipeline reproduces Ising-class finite-size scaling—order-parameter S-curves, susceptibility-peak growth with $\gamma/\nu = 1.732(8)$, and correlation-length crossings giving $\nu = 1.04(8)$, consistent with the exact Ising values, once a cluster sampler controls the critical slowing down of local updates—validating the training distributions. The bipartite structure naturally accommodates multiple field types (scalar, gauge, fermion) at the same lattice site, providing a direct path to gauge theories and lattice quantum chromodynamics (QCD) without architectural changes.

I. INTRODUCTION

Lattice quantum field theory is the primary non-perturbative tool for studying strongly coupled quantum systems, underpinning our quantitative understanding of hadron physics through lattice quantum chromodynamics (QCD) [1]. Despite decades of algorithmic advances, lattice calculations face persistent computational bottlenecks: critical slowing down near phase transitions, the fermion sign problem at finite baryon density, and the inability to directly simulate real-time dynamics in Minkowski spacetime.

Machine learning methods have shown promise in addressing several of these challenges. Normalizing flows can accelerate sampling by learning the Boltzmann distribution [2, 3], while neural networks can learn lattice observables directly from field configurations—via gauge-equivariant convolutional architectures [4] or graph neural networks (GNNs) [5]. However, existing GNN approaches treat the lattice as a *homogeneous graph*, embedding field values (scalar, gauge links, spinor components) as features on spacetime nodes. This conflates two fundamentally different objects: the geometry of spacetime and the dynamical content of quantum fields.

In continuum QFT, the action functional separates geometry from matter:

$$S[\phi, g] = \int d^d x \sqrt{g} \left[\tfrac{1}{2} g^{\mu\nu} \partial_\mu \phi \, \partial_\nu \phi + V(\phi)\right], \qquad (1)$$

where the metric $g_{\mu\nu}$ encodes spacetime geometry and ϕ encodes field content. These couple through the action but are distinct mathematical objects living in different spaces.

We propose a *heterogeneous bipartite graph* architecture built around this separation on the lattice. Spacetime nodes encode lattice geometry (coordinates, spacing, boundary conditions), while field nodes encode dynamical degrees of freedom (scalar values, gauge link matrices, spinor components). The two layers communicate through typed "inhabits" edges and three-stage message passing. This design has several advantages:

1. **Natural multi-field handling:** Multiple field types at the same site are distinct nodes with separate encoders, avoiding feature concatenation.
2. **Dynamic geometry:** Spacetime node positions can be made learnable, enabling adaptive mesh refinement.
3. **Extensibility:** Adding gauge or fermion fields requires only new encoder modules, not architectural changes.
4. **Physical interpretability:** The bipartite structure preserves the geometry–matter distinction through the entire network.

We benchmark this architecture on two-dimensional scalar ϕ^4 theory, the simplest interacting field theory with a well-understood phase transition in the Ising universality class ($\nu = 1$ exactly). This provides a rigorous test: the model must learn the action functional, reproduce

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the phase transition, and yield correct finite-size scaling across lattice sizes up to $L = 128$.

A. Related Work and Novelty

Machine learning approaches to lattice field theory have developed along several directions (see Ref. [6] for a recent review). Normalizing flows learn the Boltzmann distribution for efficient sampling [2, 3, 7]. Gauge-equivariant and gauge-invariant neural networks enforce exact lattice symmetries in the network architecture [4, 8, 9]. Graph neural networks have been applied to learn field-theoretic observables [5], and relational inductive biases have proven valuable in broader physics contexts [10, 11]. In the machine-learning literature, heterogeneous graphs with typed nodes and relations are a well-developed modeling tool [12, 13], but to our knowledge they have not been applied to the geometry–field decomposition of lattice field theory.

These approaches share a common representation: field variables are attached directly to lattice sites or links, with all information embedded on a single node type in a homogeneous graph (or equivalently on a regular grid). While effective, this conflates spacetime geometry with field content—objects that are mathematically distinct in continuum QFT.

Our contribution is a *heterogeneous bipartite graph formulation* in which spacetime geometry and field degrees of freedom are represented as distinct node types connected by typed edges. This structure is loosely analogous to a fiber bundle: spacetime nodes play the role of the discrete base manifold M, each field node represents a point in the fiber over its site (so that a field configuration — the collection of all field-node values — is a discretized section), and the "inhabits" edges encode the projection $\pi : E \to M$ associating each field value to its spacetime site. The three-stage message-passing protocol (field $\to$ spacetime, spacetime $\to$ spacetime, spacetime $\to$ field) acts as a discrete analog of the field–geometry coupling encoded in the action functional. The resulting architecture naturally supports multiple field types at a single lattice site without feature concatenation and provides a modular pathway to more complex theories, including gauge fields and fermions.

II. ARCHITECTURE

A. Heterogeneous Bipartite Graph

For a scalar field theory on an $L \times L$ lattice with periodic boundary conditions, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of two node types and three edge types:

Node types:

- *Spacetime nodes* ($N = L^2$): features are coordinates (x_1, x_2).

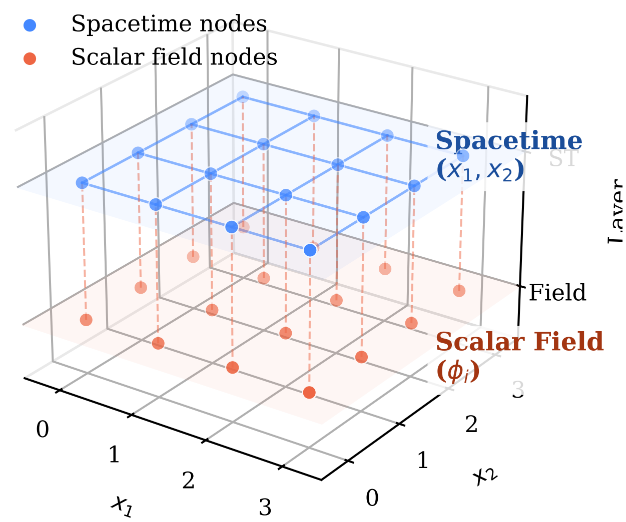

FIG. 1. 3D view of the heterogeneous bipartite graph for a 4×4 lattice. Blue spheres (upper plane): spacetime nodes with coordinate features (x_1, x_2). Orange spheres (lower plane): scalar field nodes with ϕ values. Solid blue lines: adjacency edges encoding lattice connectivity. Dashed orange lines: inhabits edges coupling each field node to its spacetime site. Reverse inhabits edges (not shown) enable bidirectional message passing.

- *Scalar field nodes* ($N = L^2$): features are the field value (ϕ_i) at each site.

Edge types:

- *Adjacent* (spacetime $\to$ spacetime): $2dN$ directed edges encoding the lattice connectivity, with displacement vectors $a\hat{e}_\mu$ as features. These encode both the direction and physical distance between neighboring sites, placing metric information on the edges where it naturally belongs.

- *Inhabits* (field $\to$ spacetime): N edges connecting each field node to its corresponding spacetime site.

- *Inhabits-inverse* (spacetime $\to$ field): reverse of the above, enabling spacetime-to-field message flow.

Figure 1 shows the resulting two-layer graph for a 4×4 lattice. For an $L \times L$ lattice, the graph has $2L^2$ nodes and $4L^2 + 2L^2 = 6L^2$ directed edges (for $d = 2$). The graph is implemented using PyTorch Geometric's [14] `HeteroData` container.

B. Model: HeteroGNN

The model consists of three stages: type-specific encoding, iterative three-stage message passing, and a physics-informed readout head.

Encoding. Each node and edge type has a dedicated multilayer perceptron (MLP) encoder that projects raw features to a shared hidden dimension H:

$$\mathbf{h}_i^{(\mathrm{st})} = \mathrm{MLP}_{\mathrm{st}}(x_i^1, x_i^2) \in \mathbb{R}^H, \tag{2}$$
$$\mathbf{h}_i^{(\mathrm{field})} = \mathrm{MLP}_{\mathrm{field}}(\phi_i) \in \mathbb{R}^H, \tag{3}$$
$$\mathbf{e}_{ij}^{(\mathrm{adj})} = \mathrm{MLP}_{\mathrm{edge}}(a\hat{e}_\mu) \in \mathbb{R}^H. \tag{4}$$

Three-stage message passing. Each message-passing block performs three sequential steps with residual connections and layer normalization:

Stage 1: Field → Spacetime. Field information is aggregated at each spacetime site via the inhabits edges:

$$\mathbf{h}_i^{(\mathrm{st})} \mathrel{+}= \mathrm{MLP}_1\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \sum_{j\in\mathcal{N}_{\mathrm{inh}}(i)} \mathbf{h}_j^{(\mathrm{field})}\right). \tag{5}$$

Stage 2: Spacetime → Spacetime. Information propagates along the lattice via adjacency edges, analogous to discretized spatial derivatives:

$$\mathbf{h}_i^{(\mathrm{st})} \mathrel{+}= \mathrm{MLP}_2\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \sum_{j\in\mathcal{N}_{\mathrm{adj}}(i)} [\mathbf{h}_j^{(\mathrm{st})} \,\|\, \mathbf{e}_{ij}]\right). \tag{6}$$

Stage 3: Spacetime → Field. Updated geometry information flows back to field nodes:

$$\mathbf{h}_i^{(\mathrm{field})} \mathrel{+}= \mathrm{MLP}_3\left(\mathbf{h}_i^{(\mathrm{field})} \,\|\, \sum_{j\in\mathcal{N}_{\mathrm{inh}}^{-1}(i)} \mathbf{h}_j^{(\mathrm{st})}\right). \tag{7}$$

Each step includes residual connections (+=) and layer normalization to mitigate over-smoothing. We stack B such blocks; with B blocks, the receptive field spans B lattice spacings.

Readout: Action head. The total Euclidean action is predicted as a sum of local action densities:

$$\hat{S}_E[\phi] = \sum_{i=1}^{N} \mathrm{MLP}_{\mathrm{action}}\left(\mathbf{h}_i^{(\mathrm{st})} \,\|\, \mathbf{h}_i^{(\mathrm{field})}\right) \cdot a^d, \tag{8}$$

where the a^d factor provides the lattice volume element, converting the discrete sum into the continuum integral $\int d^d x\, \mathcal{L}$. This factor is not merely cosmetic: ablation experiments (Sec. V A) show that removing it causes catastrophic failure at non-unit lattice spacings, as the MLP cannot efficiently learn to compensate for a scale factor spanning several orders of magnitude. The per-site decomposition mirrors the structure of the lattice action.

III. BENCHMARK: SCALAR ϕ^4 THEORY

A. Theory and Observables

We study the Euclidean action on a two-dimensional square lattice:

$$S_E[\phi] = \sum_x a^2 \left[\frac{1}{2}\sum_\mu \frac{(\phi_{x+\hat{\mu}} - \phi_x)^2}{a^2} + \frac{m^2}{2}\phi_x^2 + \lambda\phi_x^4\right], \tag{9}$$

with periodic boundary conditions. For $\lambda = 0.5$, this theory undergoes a second-order phase transition in the Ising universality class as m^2 is varied, with critical exponent $\nu = 1$ (exact, from conformal field theory [15]).

Key observables for the finite-size scaling analysis are:

- **Order parameter:** $|M| = |\langle\phi\rangle| = |V^{-1}\sum_x \phi_x|$
- **Susceptibility:** $\chi = V(\langle M^2\rangle - \langle|M|\rangle^2)$
- **Correlation length** (second-moment estimator):

$$\xi = \frac{1}{2\sin(\pi/L)}\sqrt{\frac{\tilde{G}(0)}{\tilde{G}(k_{\min})} - 1}, \tag{10}$$

where $\tilde{G}(k)$ is the Fourier-transformed connected correlator and $k_{\min} = 2\pi/L$.

We implement the correlation length estimator using the full two-dimensional FFT of the field configurations:

$$\tilde{G}(0) = \chi = V\left(\langle M^2\rangle - \langle|M|\rangle^2\right), \tag{11}$$
$$\tilde{G}(k_{\min}) = \langle|\tilde{\phi}(k_{\min}, 0)|^2\rangle/V, \tag{12}$$

which uses all spatial correlations and avoids the systematic underestimation of one-dimensional angle-averaged approaches. The $k \neq 0$ modes need no subtraction since $\langle\tilde{\phi}(k \neq 0)\rangle = 0$, but the $k = 0$ mode is subtraction-dependent: the $\langle|M|\rangle$-subtracted form degenerates in the symmetric phase (for Gaussian M, $\langle|M|\rangle^2 = (2/\pi)\langle M^2\rangle$, driving the ratio in Eq. (10) negative), while the alternative $V\,\mathrm{Var}(M)$ form is contaminated at finite volume by rare tunneling between the two ordered vacua in the broken phase. The ξ/L crossings and scaling collapse therefore use a two-momentum variant built entirely from $k \neq 0$ modes,

$$\xi^{-2} = \frac{\hat{k}_2^2\,\tilde{G}(k_2) - \hat{k}_1^2\,\tilde{G}(k_1)}{\tilde{G}(k_1) - \tilde{G}(k_2)}, \qquad \hat{k}^2 = \sum_\mu 4\sin^2(k_\mu/2), \tag{13}$$

with $k_1 = (2\pi/L)(1,0)$ (averaged with $(0,1)$) and $k_2 = (2\pi/L)(1,1)$, validated against the exact free-field propagator. The susceptibility χ itself is always the $\langle|M|\rangle$-subtracted form defined above.

B. Monte Carlo Data Generation

Field configurations are generated using the Metropolis–Hastings algorithm with single-site Gaussian proposals. For lattices of size $L \geq 32$, we employ a *checkerboard decomposition*: sites are partitioned into two sublattices by coordinate parity, and all sites on the same sublattice are updated simultaneously in a single vectorized operation. This preserves detailed balance (no two updated sites are neighbors) while achieving 10–50× speedup over sequential updates.

We generate 5000 configurations at each lattice size ($L = 8, 16, 32, 64$) for training, with $m^2 = -0.5$ and $\lambda = 0.5$; one model is trained per lattice size in Table I (Sec. IV F tests a single model across sizes). Configurations are separated by $n_{\text{sweeps}} = 10$ full sweeps to reduce autocorrelation, following 1000 thermalization sweeps. Acceptance rates are maintained near the optimal $\sim 47\%$ by tuning the proposal step size.

C. Training

The GNN is trained to minimize the mean squared error between predicted and exact Euclidean actions:

$$\mathcal{L} = \frac{1}{N_{\text{batch}}} \sum_{i=1}^{N_{\text{batch}}} \left(\hat{S}_E[\phi_i] - S_E[\phi_i] \right)^2. \tag{14}$$

Model hyperparameters: hidden dimension $H = 64$, $B = 3$ message-passing blocks, GELU activation, 2-layer MLPs in encoders; 150,337 parameters at 16×16 (150,529 for the coupling-conditioned variant of Sec. IV C). Training uses AdamW (learning rate 10^{-3}, weight decay 10^{-5}) with cosine annealing over 150 epochs, batch size 32, and an 80/20 train/validation split; seeds and all other settings are recorded in the committed run configurations (`configs/paper/`) and per-run provenance files (`results/`). Monte Carlo data generation runs on laptop CPU; training runs on a single Colab GPU (T4-class, ∼12 min per seed at 16×16, ∼63 min at 64×64).

IV. RESULTS

A. Action Prediction

Table I summarizes the action prediction results. The model achieves near-perfect correlation across lattice sizes, demonstrating that the heterogeneous bipartite architecture can faithfully represent the action functional of scalar ϕ^4 theory. At 64×64, four of five seeds reach $r \geq 0.9989$, while one converges slowly, reaching only $r = 0.910$ within the fixed budget but $r = 0.9998$ when the epoch budget (and hence the cosine annealing schedule) is doubled — slow convergence, not a failure mode.

TABLE I. Action prediction accuracy across lattice sizes. Pearson correlation r and mean relative error are computed on held-out validation sets over 5 seeds. We report $1 - r$ (smaller is better) since correlations are near unity; parenthetical uncertainties are seed-to-seed standard deviations.

Lattice	Sites	$1 - r$	Rel. Error
8×8	64	$7.4(1.0) \times 10^{-6}$	$0.05 \pm 0.00\%$
16×16	256	$2.4(0.5) \times 10^{-5}$	$0.05 \pm 0.00\%$
32×32	1024	$1.2(0.2) \times 10^{-4}$	$0.04 \pm 0.00\%$
64×64	4096	$1.9(3.6) \times 10^{-2}$	$0.23 \pm 0.32\%$

TABLE II. Architecture comparison on 16×16 lattice ($m^2 = -0.5$, $\lambda = 0.5$). All models trained for 150 epochs on the same data split over 5 seeds. We report $1 - r$ (smaller is better) since correlations are near unity; parenthetical uncertainties are seed-to-seed standard deviations.

Model	Params	$1 - r$	Rel. Error
HeteroGNN (ours)	150,337	$2.4(0.5) \times 10^{-5}$	$0.05 \pm 0.00\%$
Homogeneous GNN	67,073	$1.4(0.3) \times 10^{-4}$	$0.11 \pm 0.01\%$
Lattice CNN	29,153	$5.0(2.8) \times 10^{-3}$	$0.67 \pm 0.25\%$
Lattice CNN (matched)	146,233	$4.7(2.2) \times 10^{-4}$	$0.22 \pm 0.05\%$
MLP	197,633	$6.6(0.1) \times 10^{-1}$	$7.37 \pm 0.03\%$

We report the unfiltered 150-epoch statistics, which inflate the quoted standard deviation; the same seed protocol at $L \leq 32$ shows seed-to-seed spreads of 2×10^{-5} or less. The mild decrease in typical accuracy with lattice size is expected: the graph has 16× more nodes at 64×64, and the receptive field of $B = 3$ message-passing blocks covers a smaller fraction of the lattice.

B. Architecture Comparison

To isolate the contribution of the bipartite architecture, we compare against three baselines trained on the same 16×16 data with matched training protocols (150 epochs, AdamW, cosine annealing):

- **Homogeneous GNN:** Field values concatenated with spacetime coordinates on a single node type; same message-passing depth ($B = 3$) and hidden dimension ($H = 64$), with a skip connection from the initial encoding to the readout to prevent oversmoothing of the field signal.
- **Lattice convolutional neural network (CNN):** Periodic-padded 2D convolutions treating $\phi(x)$ as a single-channel image (4 layers, 32 channels), plus a *parameter-matched* variant (4 layers, 72 channels, 146k parameters ≈ the HeteroGNN's 150k) to control for model capacity.
- **MLP:** Flattened field configuration fed through a 3-layer feedforward network ($H = 256$).

Table II shows the results. The most significant finding is a *structural failure mode* in the homogeneous GNN: without an explicit skip connection from its initial encoding to the readout head, message passing progressively destroys the field signal. Figure 2 quantifies this with a depth ablation (three seeds per point): the no-skip homogeneous model is intact at $B = 1$ ($r = 0.9995$), becomes bimodal across seeds at $B = 2$–3 ($r = 0.43$–0.46 ± 0.4 — individual seeds either partially retain or lose the signal), and collapses completely by $B = 4$–6 ($r = 0.07 \pm 0.09$ and 0.009 ± 0.013). The skip connection is an architectural workaround, required to preserve the original field signal because field values and geometry features share a single node representation that is progressively smoothed during message passing. The bipartite architecture avoids this failure mode by construction: field nodes maintain a separate representation throughout the network, and Fig. 2 shows both the HeteroGNN and the skip-equipped homogeneous model holding $r > 0.999$ at every depth tested. Note that the per-stage residual connections in the HeteroGNN (Sec. II B) serve a different purpose: they stabilize gradient flow within each message-passing block, whereas the homogeneous skip connection bypasses message passing entirely to rescue a signal that would otherwise be destroyed.

Beyond architectural stability, all spatially structured models achieve $r > 0.99$, confirming that the ϕ^4 action is learnable as a local functional. The HeteroGNN achieves the lowest relative error (0.05%), roughly half that of the homogeneous GNN with its skip connection (0.11%). Matching the CNN's parameter count to the HeteroGNN's (146k vs. 150k) narrows but does not close the gap: the matched CNN reaches $r = 0.9995$ and 0.22% relative error, a 4× larger error than the HeteroGNN at equal capacity, indicating the advantage is architectural rather than a parameter-budget artifact. The MLP baseline fails without any spatial structure ($r = 0.34$), confirming that the lattice topology is essential. Seed-to-seed variation is negligible for the graph models ($\sigma_r \lesssim 3 \times 10^{-5}$) and modest for the CNNs, so the ordering in Table II is stable.

C. Coupling Generalization

To test whether the model learns a transferable action *family* rather than a single action landscape, the coupling constants are made model inputs: field-node features become $[\phi_i, m^2, \lambda]$ and the readout MLP is conditioned on (m^2, λ) (adding 192 parameters). A single model is then trained *jointly* on alternating points of a 13-point m^2 grid spanning $[-2.9, -0.3]$ at $\lambda = 0.5$ — both phases and the critical region — with 3000 configurations per point (thermalization ≥ 2000 sweeps; separations scaled by the measured τ_{int}, up to 72 sweeps near criticality). The held-out couplings interleave the training points (interpolation test), and both grid endpoints are excluded from training (extrapolation test).

TABLE III. Generalization across coupling values for a single coupling-conditioned model trained jointly on the †-marked m^2 points and evaluated on 1000 held-out configurations per coupling ($*$ = extrapolation beyond the training grid), over 5 seeds. We report $1-r$ (smaller is better) since correlations are near unity; parenthetical uncertainties are seed-to-seed standard deviations. Relative error is inflated where the action distribution crosses zero (near $m^2 \approx -2.5$); Pearson r is the meaningful metric there.

m^2	$1-r$	Rel. Error
-2.9^*	$2.8(1.2) \times 10^{-3}$	$19.52 \pm 2.65\%$
$-2.6833^\dagger$	$6.9(0.5) \times 10^{-6}$	$0.07 \pm 0.00\%$
-2.4667	$5.2(1.0) \times 10^{-5}$	$10.34 \pm 1.06\%$
$-2.25^\dagger$	$7.5(0.8) \times 10^{-6}$	$5.58 \pm 0.75\%$
-2.0333	$6.4(0.9) \times 10^{-5}$	$0.46 \pm 0.04\%$
$-1.8167^\dagger$	$1.8(0.1) \times 10^{-5}$	$0.08 \pm 0.00\%$
-1.6	$2.3(0.2) \times 10^{-5}$	$0.07 \pm 0.01\%$
$-1.3833^\dagger$	$1.7(0.1) \times 10^{-5}$	$0.05 \pm 0.00\%$
-1.1667	$1.9(0.2) \times 10^{-5}$	$0.06 \pm 0.01\%$
$-0.95^\dagger$	$1.4(0.1) \times 10^{-5}$	$0.04 \pm 0.00\%$
-0.7333	$2.0(0.3) \times 10^{-5}$	$0.08 \pm 0.03\%$
$-0.5167^\dagger$	$1.4(0.1) \times 10^{-5}$	$0.04 \pm 0.00\%$
-0.3^*	$6.2(3.2) \times 10^{-5}$	$0.55 \pm 0.34\%$

In an earlier version of this experiment, a model trained at a single coupling with no coupling input was evaluated against target actions computed at each new m^2; since no function of ϕ alone can match a family of action functionals indexed by a hidden parameter, that protocol measured a partially unlearnable label shift on top of the distribution shift, and the apparent degradation was severe (e.g. $r = 0.58$ at $m^2 = -2.0$). Table III reports the corrected protocol.

Table III shows that with the couplings as inputs, the single model interpolates essentially perfectly: every held-out interpolation coupling reaches $r \geq 0.9999$, *including* $m^2 = -2.47$ in the critical region where the old protocol appeared to fail. Extrapolation beyond the training grid degrades only at the deep broken-phase endpoint ($m^2 = -2.9$: $r = 0.9972 \pm 0.0012$), while the symmetric-phase endpoint remains at $r = 0.9999$. Because the action is a strictly local functional of the field, its prediction does not suffer near criticality once the label family is identified — long-range physics enters the ensemble *distribution*, not the action's functional form.

D. Phase Transition and Finite-Size Scaling of the Training Ensembles

This and the following subsection characterize the *Monte Carlo ensembles* used throughout this work: they validate the data-generation pipeline against known Ising-universality results, rather than testing the network itself. We map the phase transition of the two-dimensional ϕ^4 theory at $\lambda = 0.5$ by a coupling sweep in m^2. Because the observables of interest live at

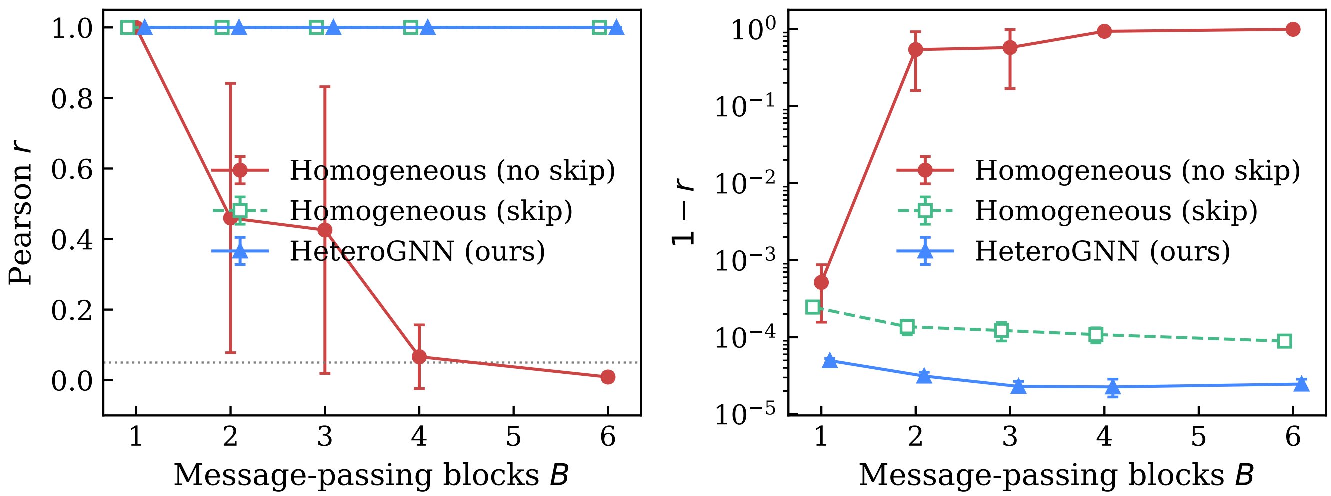

FIG. 2. Depth ablation at 16×16 (mean $\pm$ std over 3 seeds). **Left:** Pearson r vs. number of message-passing blocks B. The homogeneous GNN without a readout skip connection degrades monotonically with depth and collapses below $r = 0.05$ (dotted line) by $B = 6$; the large error bars at $B = 2$–3 reflect bimodal seed-to-seed outcomes. **Right:** $1 - r$ on a log scale, resolving the persistent gap between the HeteroGNN and the skip-equipped homogeneous baseline at every depth.

the critical point—exactly where the local Metropolis update suffers critical slowing down—we generate the finite-size-scaling ensembles with a Wolff/Brower–Tamayo embedded-cluster sampler [16, 17]: single-cluster $\mathbb{Z}_2$ sign flips, whose dynamic exponent is $z \approx 0.25$ for this universality class, interleaved with local sweeps that update the field magnitude. This holds the integrated autocorrelation time to $\mathcal{O}(1)$ across the entire scan and lets us reach 7 sizes up to $L = 128$. We retain the local-Metropolis ensembles at the same couplings as a direct measurement of the sampling bias they carry.

Figure 3 shows the three canonical finite-size-scaling observables. The qualitative picture—an order-parameter S-curve sharpening with L, a growing susceptibility peak, and crossing ξ/L curves—is reproduced by both samplers; the local ensembles, however, are quantitatively biased by critical slowing down. Table IV lists the integrated autocorrelation time of $|M|$ at each pseudocritical point in single sweeps: for the local update it grows from 65 at $L = 16$ to 185 at $L = 64$ ($\tau \sim \xi^z$, $z \approx 2$), while the cluster hybrid holds $\tau_{\text{int}} \approx 8$–9 independent of L—a decorrelation speed-up that reaches $21\times$ by $L = 64$ and makes $L = 96, 128$ practical at all.

The resulting bias in the exponent is direct. For the local ensembles the effective slope of $\ln \chi_{\max}$ versus $\ln L$ falls monotonically with volume, from 1.64 ($L = 16$–24) to 1.42 ($L = 48$–64) as the largest lattices lose effective statistics, and a global fit gives the downward-biased $\gamma/\nu = 1.60(3)$. The cluster effective slope shows no such trend. Locating the peaks by quadratic fits to dense scans centered on each maximum (grid width $\propto L^{-1}$, so every fit spans the same fraction of its peak and $\chi^2/\text{dof} \sim 1$), a weighted fit of $\ln \chi_{\max}$ versus $\ln L$ over all 7 sizes gives

$$\gamma/\nu = 1.732(8), \qquad (15)$$

TABLE IV. Susceptibility peak height $\chi_{\max}$ and $|M|$ integrated autocorrelation time (single sweeps, at the pseudocritical point) for the local-Metropolis and cluster samplers. The local τ_{int} grows as ξ^z ($z \approx 2$); the cluster's stays $\mathcal{O}(1)$, and the local sampler is impractical beyond $L = 64$ at these statistics.

	$\chi_{\max}$		$\tau_{\text{int}}(\lvert M\rvert)$		
L	local	cluster	local	cluster	speed-up
16	9.6(2)	8.68(12)	65	8.3	8
24	18.7(6)	18.2(2)	104	12.8	8
32	29.7(10)	29.0(4)	92	8.3	11
48	56.5(18)	58.9(7)	141	8.6	16
64	85(4)	96.9(19)	185	8.7	21
96	–	198(3)	–	8.7	–
128	–	319(5)	–	8.7	–

consistent with the exact two-dimensional Ising value $\gamma/\nu = 1.75$. A corrections-to-scaling fit $\chi_{\max} = A\,L^{\gamma/\nu}(1 + b\,L^{-\omega})$ with $\omega = 2$, and an Akaike-weighted average over fit ranges [18], give 1.720(16) and 1.731(16) respectively, consistent with the naive fit.

E. Critical Exponent Extraction

The correlation-length exponent and the critical coupling follow from the approach of the pseudocritical points to the infinite-volume limit, $m^2_{\text{peak}}(L) = m^2_c - a\,L^{-1/\nu}$. A three-parameter fit to the 7 cluster peak locations gives

$$\nu = 1.04(8), \qquad m^2_c = -2.217(3) \quad (\lambda = 0.5), \qquad (16)$$

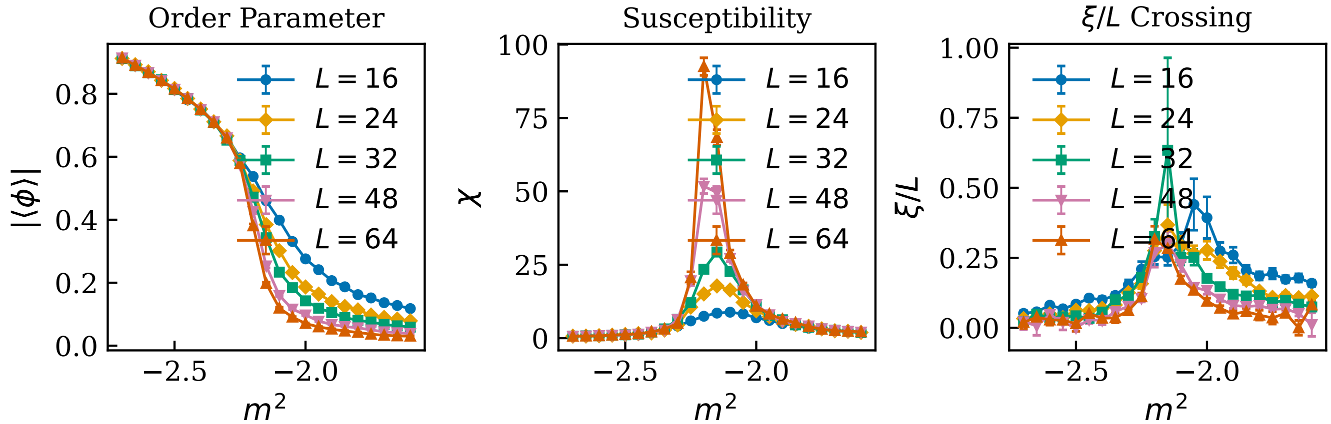

FIG. 3. Finite-size scaling of the cluster ensembles at $\lambda = 0.5$. **Left:** order parameter $|M|$, sharpening with L. **Center:** susceptibility χ, with peaks growing as $\chi_{\max} \sim L^{\gamma/\nu}$. **Right:** correlation-length ratio ξ/L, whose curves cross near $m_c^2 = -2.217(3)$. Error bars are binned jackknife errors with bin sizes set by the measured τ_{int}; per-L markers are colorblind-distinct.

with $\chi^2/\mathrm{dof} = 1.6$, consistent with the exact Ising $\nu = 1$. We quote this pseudocritical-shift fit rather than a ξ/L data collapse because the naive collapse-quality metric has no proper interior minimum on these data.[1]

Two points frame these results. First, this is a validation of the training ensembles, not a precision lattice study: dedicated determinations of the two-dimensional ϕ^4 critical line [19, 20] reach far higher precision and use a different coupling normalization (continuum-limit ratios $f = \lambda/\mu_c^2$ rather than a fixed bare λ at finite spacing), so we do not attempt a quantitative comparison. Second—and this is why we belabor the sampling—the exercise makes concrete the bottleneck that motivates the broader program. The cluster algorithm that removes the bias here exists only because the scalar theory has a simple $\mathbb{Z}_2$ order parameter; no comparable algorithm is known for the gauge and fermion theories targeted in the subsequent phases, where critical slowing down (and, with dynamical fermions, the cost of the determinant) is the dominant obstacle to generating training data. A learned surrogate for the action—the architecture developed here—is one route into precisely that regime, where exact cluster methods are unavailable.

F. Size Transfer

The models above are trained one-per-lattice-size. To test whether a single learned action functional transfers across volumes, we train at 16×16 only and evaluate at $L \in \{8, 32, 64\}$ without any retraining, for two spacetime-feature variants: absolute coordinates (as in the rest of this paper) and a coordinate-free variant in which spacetime nodes carry a constant feature and all geometry resides in the displacement edge features (restoring exact translation invariance).

TABLE V. Size transfer: a single model trained at 16×16 (†, validation split) evaluated at other lattice sizes without retraining, over 3 seeds. We report $1 - r$ (smaller is better) since correlations are near unity; parenthetical uncertainties are seed-to-seed standard deviations.

Variant	Eval. lattice	$1 - r$	Rel. Error
Coordinates	8×8	$2.8(0.1) \times 10^{-5}$	$0.10 \pm 0.00\%$
	$16 \times 16^{\dagger}$	$2.4(0.4) \times 10^{-5}$	$0.05 \pm 0.00\%$
	32×32	$2.4(0.3) \times 10^{-5}$	$0.03 \pm 0.00\%$
	64×64	$2.0(0.3) \times 10^{-5}$	$0.02 \pm 0.01\%$
Coordinate-free	8×8	$1.5(0.2) \times 10^{-5}$	$0.08 \pm 0.01\%$
	$16 \times 16^{\dagger}$	$1.5(0.3) \times 10^{-5}$	$0.04 \pm 0.00\%$
	32×32	$1.6(0.2) \times 10^{-5}$	$0.02 \pm 0.00\%$
	64×64	$1.4(0.2) \times 10^{-5}$	$0.01 \pm 0.00\%$

Table V shows that the learned functional transfers essentially perfectly in both directions — down to 8×8 and up to 64×64, a $64\times$ change in volume — with $r = 1.0000$ at every size for both variants. Relative error *decreases* with lattice size (from 0.10% at $L = 8$ to 0.01–0.02% at $L = 64$), consistent with an accurate learned per-site action density whose fluctuations average out in the extensive sum. The per-site readout (Eq. (8)) is what makes this possible: the same local functional applies at any volume. Notably, even the absolute-coordinate variant transfers, indicating the model learns to ignore the physically irrelevant coordinate inputs; the coordinate-free variant is nevertheless preferable on principle (exact translation invariance) and attains slightly lower relative error at every size.

[1] The same fit on the local ensembles gives $\nu = 1.0(5)$: the central value agrees, but the local peak positions scatter under critical slowing down ($\chi^2/\mathrm{dof} \approx 4$), inflating the error roughly sixfold.

V. DISCUSSION

A. Advantages of the Bipartite Architecture

The heterogeneous bipartite graph offers several structural advantages over homogeneous approaches:

Multi-field extensibility. In the homogeneous approach, adding a gauge field $U_\mu(x) \in \mathrm{SU}(N)$ requires concatenating its matrix elements with the scalar field value at each node, breaking the natural group structure. In our architecture, gauge fields become a separate node type with their own encoder, connected to spacetime via typed edges. The three-stage message-passing blocks handle arbitrary numbers of field types without modification.

Physical interpretability. The action readout (Eq. (8)) decomposes the total action into per-site contributions $\hat{s}(x)$ that can be directly compared with the lattice action density, enabling local error analysis.

Metric information placement. The lattice spacing a is a metric quantity—it defines the physical distance between neighboring sites. We encode it on adjacency edges as displacement vectors $a\hat{e}_\mu$ rather than as a node feature, reflecting the fact that distances are properties of connections, not of individual points. To validate this choice, we tested four architecture variants (lattice spacing in nodes vs. edges, with and without the explicit a^d volume factor in Eq. (8)) across lattice sizes 8×8 through 64×64 at varying lattice spacings $a = 1/L$. Two findings emerged. First, the a^d volume scaling is essential: without it, performance collapses as a decreases (Pearson r drops from > 0.999 to -0.26 at 64×64 with $a = 1/64$), because the MLP readout cannot learn to compensate for a scale factor spanning four orders of magnitude. Second, the choice of whether a resides on nodes or edges has negligible effect on accuracy ($r > 0.999$ for both at all sizes), confirming that the placement is a design choice rather than a performance constraint. We choose edges because the metric naturally lives on links, and this design extends cleanly to gauge theories where link variables $U_\mu(x)$ share the same edge representation.

Over-smoothing mitigation. By separating field and spacetime representations, the architecture avoids mixing all information into a single node type during message passing. Residual connections and layer normalization further stabilize deep stacking.

Interpretation as a learned effective action. The action readout (Eq. (8)) decomposes the total action into per-site contributions, and the finite receptive field of B message-passing blocks limits each contribution to information within Ba lattice spacings. The model therefore admits an interpretation as a *local effective action* operator—analogous to a Wilsonian effective action in which modes above a momentum cutoff $\Lambda \sim 1/(Ba)$ have been integrated out. In this view, the message-passing depth B plays the role of an implicit renormalization scale, controlling the resolution at which the theory is represented. This interpretation explains both the model's success on the local ϕ^4 action and its degradation near criticality, where the relevant physics involves correlations at scales $\xi \gg Ba$ that lie below the effective cutoff.

B. Limitations

Critical slowing down. The Metropolis sampler's autocorrelation time diverges at the critical point, limiting our ability to precisely determine ν. This is a sampling limitation, not an architectural one.

Receptive field and locality scale. With $B = 3$ blocks, the model's receptive field spans $\ell = Ba = 3a$, defining a finite *locality scale* for the learned operator. This is sufficient for the ϕ^4 action, which is a strictly local functional of the field and its nearest-neighbor differences — and indeed action prediction remains accurate even at couplings near criticality once the couplings are model inputs (Sec. IV C): the diverging correlation length changes the ensemble *distribution*, not the action's local functional form. The locality scale becomes a genuine restriction for observables that are themselves nonlocal — correlators at separations $\gg \ell$, Wilson loops of large area, or spectral quantities — which the network cannot represent regardless of coupling. Increasing B enlarges the locality scale but risks over-smoothing, where distinct node representations collapse toward a common value (measured directly in Fig. 2 for the homogeneous baseline). For theories requiring larger locality scales, possible mitigations include multi-scale pooling layers, skip connections across non-adjacent blocks, or attention mechanisms that allow selective long-range communication.

Lattice symmetries. The current architecture does not explicitly enforce discrete lattice symmetries (translations, C_4 rotations, reflections). Instead, these symmetries are learned implicitly from the data, which is generated with periodic boundary conditions that respect the full symmetry group. This is a deliberate design choice, not a limitation: gauge-equivariant architectures [4, 8] — including, recently, gauge-equivariant message passing on graphs [21] — achieve exact symmetry by constraining the network weights and message functions, but this rigidity restricts them to the symmetry group built in at design time. Our approach prioritizes *architectural flexibility*—the same model structure handles irregular meshes, non-periodic boundaries, and adaptive lattice refinements without modification. For the multi-theory extensibility that motivates this work (gauge fields, fermions, curved spacetimes), this flexibility is essential. Phase II of this program (Sec. VI) will quantify the trade-off directly by measuring the *learned* gauge-invariance error of the bipartite architecture against exactly equivariant designs such as Ref. [21].

VI. FUTURE DIRECTIONS

The bipartite architecture is designed as a foundation for progressively more complex theories:

Phase II: U(1) gauge theory with fermions. Gauge link variables $U_\mu(x) = e^{iA_\mu(x)} \in \mathrm{U}(1)$ and Dirac spinors $\psi(x) \in \mathbb{C}^2$ are added as new field node types with dedicated encoders. The three-stage message passing naturally handles the tripartite field–gauge–spacetime coupling. This phase will benchmark against the Schwinger model (QED in 1+1D), which has exact solutions for comparison.

Phase III: SU(3) Yang–Mills in 4D. In four dimensions, with link encoders structured to respect the SU(3) group, the target is to reproduce established lattice QCD benchmarks such as the string tension and glueball spectrum. Connecting the learned Euclidean representation to real-time Minkowski physics is a longer-term and deliberately speculative direction.

VII. CONCLUSION

The central result of this work is that representing lattice field theory as a *heterogeneous* graph—with spacetime geometry and field degrees of freedom as distinct node types—eliminates a structural failure mode of the homogeneous graph representations used previously, in which message passing destroys the field signal without ad hoc skip connections. This geometry–field separation, inspired by the geometry–matter distinction fundamental to continuum QFT and, to our knowledge, absent from prior lattice-QFT neural architectures [4, 8, 9], preserves a physically interpretable local action decomposition. On two-dimensional scalar ϕ^4 theory, the model achieves near-perfect action prediction and transfers across a 64× range of lattice volumes without retraining; the underlying Monte Carlo pipeline reproduces the expected Ising-class finite-size scaling—$\gamma/\nu = 1.732(8)$ and $\nu = 1.04(8)$, consistent with the exact Ising values, once a cluster sampler removes the critical-slowing-down bias of local updates—validating the ensembles the model is trained on. Comparison against homogeneous GNN, CNN, and MLP baselines demonstrates the viability of the bipartite representation for lattice field theory.

The architecture's key contribution is not improved accuracy on this benchmark but its *extensibility and physical alignment*: the same graph structure and message-passing protocol accommodate scalar, gauge, and fermion fields through modular encoder additions, while preserving the geometry–field distinction throughout the computation. Metric information (lattice spacing) is encoded on adjacency edges as displacement vectors, placing it where distances physically reside and enabling clean integration with gauge link variables in future phases. This provides a unified framework for progressing from toy models to physically relevant theories, with lattice QCD as the ultimate target.

Code and data are available at `https://github.com/crosis19/qft-graph`.

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DATA AVAILABILITY

The analysis-level data supporting this study—Monte Carlo observables, finite-size-scaling fit inputs and outputs, and training metrics—and all code required to regenerate every figure, table, and quoted number are openly available at `https://github.com/crosis19/qft-graph`. Raw Monte Carlo field configurations and model checkpoints are not deposited owing to their size, but are exactly regenerable from the committed configuration files and seeds.

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